Graduate Analysis 1: midsem exam

Aditya Padiyar

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Definition 1. *A sequence of functions f_n converges almost uniformly to f if for every $\epsilon > 0$ there is a set A_ϵ with $\mu(A_\epsilon) < \epsilon$ such that f_n converges uniformly to f on $(A_\epsilon)^c$*

Definition 2. *A sequence $f_n \in L^1$ is **uniformly integrable** if for all $\epsilon > 0$ there is $\delta >$ such that $\int_A |f_n| < \epsilon$ for all sets A with $\mu(A) < \delta$*

Claim 1. *The sum of two uniformly integrable sequences is uniformly integrable*

Claim 2. *On a finite measure space (X, Σ, μ), convergence almost everywhere is equivalent to convergence almost uniformly*

Proof. Consider the following:

- See Egorov's theorem in Axler for a proof of the forward implication
- Let $B \in \Sigma$ denote the points where convergence does not occur. Then $B \subseteq A_\epsilon$ which implies that $\mu(B) = 0$

□

Claim 3. *Suppose $f_n^0, f_n^1, f_n^2, f_0, f_1, f_2$ are functions such that*

- $f_n^i, f_0, f_2 \in L^1$
- $f_n^0 \leq f_n^1 \leq f_n^2$
- $f_n^i \to f_i$
- $\int f_n^0 \to \int f_0$ *and* $\int f_n^2 \to \int f_2$

Then we have

- $f_1 \in L^1$
- $\int f_n^1 \to \int f_1$

Proof. Consider the following:

- Taking limits we have $f_0 \leq f_1 \leq f_2$. Therefore $|f_1| \leq |f_0| + |f_2|$ which shows that $f_1 \in L^1$
- Define the following:
  - $g_n = f_n^1 - f_n^0$
  - $h_n = f_n^2 - f_n^1$
  - $g = f_1 - f_0$
  - $h = f_2 - f_1$
- By Fatou's lemma $\int g \leq \liminf \int g_n$ and $\int h \leq \liminf \int h_n$
- We have

$$\begin{aligned} \limsup \int g_n &= \limsup \int (g_n + h_n - h_n) \\ &= \int f_2 - \int f_0 - \liminf \int h_n \leq \int g \end{aligned}$$

- Hence $\int g_n \to \int g$
- We have $\int f_n^1 = \int g_n + \int f_n^0 \to \int g + \int f_0 = \int f_1$

□

Claim 4. *Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is measurable and satisfies:*

- *$[x \to f(x_0, x)] \in L^1$ for all x_0*
- *$g(x, y) = \partial_x f(x, y)$ exists everywhere*
- *There is $h \in L^1$ on $\mathbb{R}$ such that $|g(x, y)| \leq h(y)$*

Let $\phi(x) = \int f(x, y)dy$. Then $\phi'(x) = \int g(x, y)dy$

Proof. Consider the following:

- Fix $x \in \mathbb{R}$ and define $q(t, y) = \frac{f(x+t,y)-f(x,y)}{t}$ for $t \neq 0$
- By the mean value theorem $q(t, y) = g(\alpha(t, y), y)$
- Pick $t_n \neq 0$ such that $t_n \to 0$
- We have $\frac{\phi(x+t_n)-\phi(x)}{t_n} = \int q(t_n, y)dy$
- We have $q(t_n, y) \to g(x, y)$ for every y
- By DCT $\phi'(x) = \int g(x, y)dy$

□

Claim 5. *Suppose X is a finite measure space. If $f_n : X \to \mathbb{R}$ are uniformly integrable and converge pointwise to $f : X \to \mathbb{R}$, then $\int |f_n - f| \to 0$*

Proof. Consider the following:

- We may assume $f = 0$ by 3.28 of Axler and 1
- Let $\alpha > 0$ and δ_α be such that $\mu(A) < \delta_\alpha$ makes $\int_A |f_n| < \frac{\alpha}{2}$
- Choose A_α with $\mu(A) < \delta_\alpha$ such that f_n converges uniformly to 0 on $(A_\alpha)^c$
- Then

$$\int |f_n| = \int_{A_\alpha} |f_n| + \int_{(A_\alpha)^c} |f_n| \leq \frac{\alpha}{2} + \int_{(A_\alpha)^c} |f_n| \to \frac{\alpha}{2}$$

□

Claim 6. *Suppose X is a random variable with finite expectation and $F \subseteq \mathbb{R}$ is closed. Suppose that whenever $P(E) > 0$ we have $\frac{\mathbb{E}[X1_E]}{P(E)} \in F$. Then $P(X \in F) = 1$*

Proof. Consider the following:

- Suppose $P(X \notin F) > 0$
- Then there is a compact interval $[a, b] \subseteq F^c$ such that $P(X \in [a, b]) > 0$
- Let E be the event $\{w : X(w) \in [a, b]\}$
- We have $\frac{\mathbb{E}[X1_E]}{P(E)} \in [a, b]$ which is a contradiction

□

Claim 7. *Define the Cantor set as in 2.74 of Axler. Define $f = \sum_{n=1}^{\infty} n1_{G_n}$. Then we have*

- *f is measurable*
- *$\int f = 3$*

Proof. Define $f_k = \sum_{n=1}^{k} n1_{G_n}$. Then f_k converges to f pointwise. This makes f measurable. By the monotone convergence theorem,

$$\int f = \sum_{n=1}^{\infty} n\lambda(G_n)$$

Since $\lambda(G_n) = \frac{1}{2}(\frac{2}{3})^n$, the result follows □

Claim 8. *Suppose we have the following:*

- $E \subseteq [0,1] \times [0,1]$ *is measurable*
- $A = \{x : \lambda(E_x) \leq \frac{1}{2}\}$ *and* $B = \{y : \lambda(E^y) = 1\}$
- $\lambda(A) = 1$

Then we have $\lambda(B) \leq \frac{1}{2}$

Proof. Consider the following:

- By Tonelli's theorem, $\lambda^2(E) = \int_0^1 \lambda(E_x)dx = \int_A \lambda(E_x) \leq \frac{1}{2}$
- We also have $\lambda^2(E) = \int_0^1 \lambda(E_y)dy \geq \int_B \lambda(E_y)dy = \lambda(B)$

□